

Retail Sales Performance Analysis – Microsoft Excel Project

Week 1 Portfolio Project — Retail & E-commerce Industry

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Retail Sales Analysis

Executive Summary

JPT Retail Stores Ltd achieved total revenue of \$1,787,419.92, of which profit was \$453,870.44 from 1,825 transactions, with a good and consistent gross margin of roughly 25.4% in the fiscal year 2023. Product level results were consistent across 5 categories of Products (Books, Electronics, Clothing, Home Goods, Sports) with no major seasonal peaks and troughs during the year. The obvious area that requires management focus is the Eastern sales region which was approximately 16% under the best performing region, while also needing to improve the data quality within their area (by removing zero-value sales entries, and correctly labelling product categories) in order to maintain the accuracy of reporting.

Business Questions

This report answers six core business questions: total sales value, average sales value, the most frequent product category, the highest and lowest recorded sales, and what overall sales performance tells management. Full answers with supporting figures are provided in Section 3.

Charts

Four charts support the analysis: Total Sales by Category, Total Sales by Region, the quantity category, Total sum of sales and finally pie chart of Sales and profit. These are presented with commentary in Section 4 (Visual Analysis).

Business Insights

Notable Findings are: No one product category is outperforming sales. It appears that I have healthy, diversified product. Sales have performed the highest in both West and South and performed low in East which is behind

roughly by \$77,000. Monthly sales have relatively stable with sales on Jul has been highest and lowest in Feb and there exist data entry issues which lead to recording 45 records with value of \$0 and 7 record with incorrect category, under reporting total sales performance.

Recommendations

Four are offered to management; they include balancing investments across all 5 product segments, undertaking a quality of data audit before the next reporting period, targeting Book and Electronic segments via aggressive advertising, and probing sales deficit in the Eastern region. Supporting facts for each of these is laid out in Section 5.

1. Project Overview

This report presents an analysis of JPT Retail Stores Ltd.'s retail sales data, conducted as a Week 1 portfolio project for the Professional Certificate in Data Analysis. As Junior Data Analyst, I was asked by the Sales Director to summarize company sales performance and provide evidence-based insights to support management decision-making. The dataset covers 1,821 sales transactions recorded across the 2023 financial year, spanning five product categories (Electronics, Clothing, Home Goods, Sports, and Books) and four sales regions (North, South, East, West).

2. Business Metrics

The table below summarizes the key business metrics calculated from the dataset using Microsoft Excel (SUM, AVERAGE, MAX, MIN, and IF functions).

Business Metric	Value
Total Sales	\$1,787,419.92
Total Profit	\$453,870.44
Total Quantity Sold	18,298 units
Average Sales (per transaction)	\$980.48
Average Profit (per transaction)	\$248.70
Highest Sale	\$1,888.93
Lowest Sale	\$43.66
Number of Sales Records	1,821
Number of Electronics Sales	361
Number of Home Goods Sales	364
Number of Sports Sales	364

Business Metric	Value
Number of Books Sales	364

** Data quality note: 45 records show a Sales value of \$0.00 despite carrying valid Quantity and Profit figures — this is almost certainly a data entry error rather than a genuine zero-value sale. Excluding these records, the true lowest recorded sale is \$43.66. A further 7 records carry invalid or missing category labels (blank, "Null", "NaN", "NaN?") and are excluded from the category breakdown below. These anomalies should be corrected at source before the dataset is used for further analysis.*

2.1 Sales Performance by Category

Category	Total Sales	Total Profit	Records
Books	\$370,971.87	\$92,304.45	364
Electronics	\$364,090.41	\$90,439.70	361
Clothing	\$350,981.82	\$90,082.12	365
Home Goods	\$347,493.83	\$87,447.82	364
Sports	\$346,685.06	\$92,065.22	364

3. Answers to Business Questions

1. What is the total sales value?

The total sales value for the period was \$1,787,419.92 across 1,821 recorded transactions.

2. What is the average sales value?

The average sales value per transaction was \$980.48.

3. Which product category appears most frequently?

Clothes seem to be a popular purchase item, 365 transactions. The category range was quite close from each other (361-365 items) suggesting the store carefully diversifies its products rather than has one dominant category.

4. What was the highest sale?

The highest single sale recorded was \$1,888.93, in the Sports category.

5. What was the lowest sale?

The lowest recorded sale value was \$0.00, affecting 45 transactions. As these appear to be data entry errors rather than genuine sales, so the lowest genuine sale value was \$43.66 (Clothing category).

6. Based on your analysis, what does the sales performance tell management?

The overall sales performance looks strong and well-distributed across categories. The overall sales are \$1,787,419.92, with an average profit margin around 25.4%. No one category seems to be overperforming

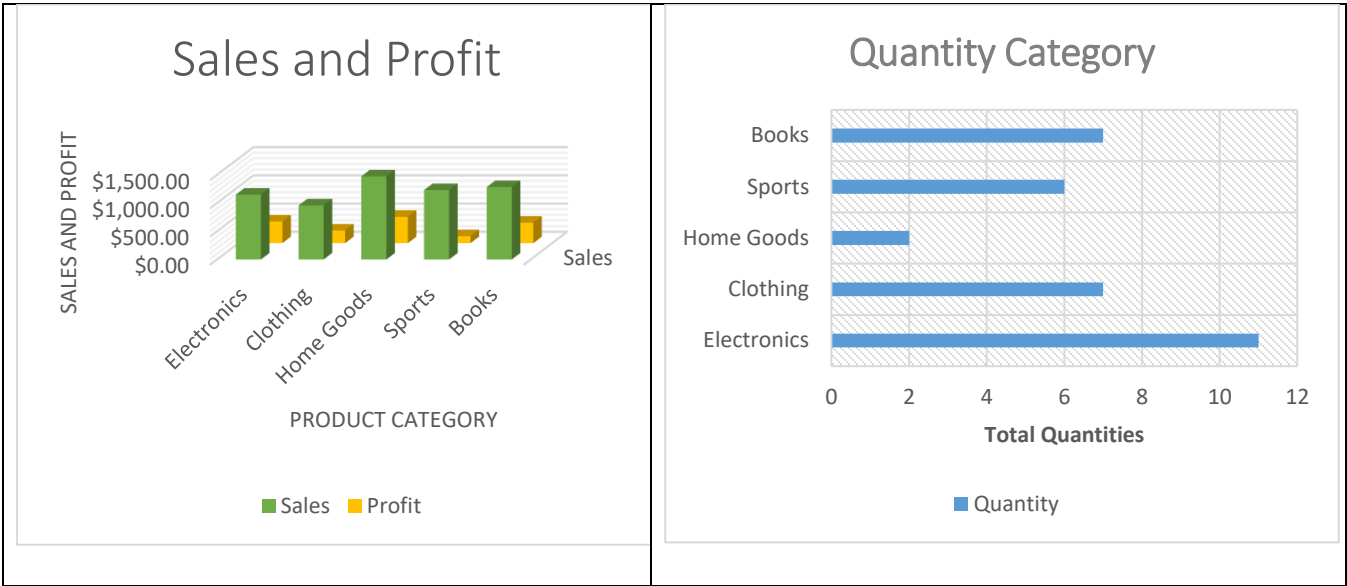
significantly over others, Books are leading slightly in the gross revenue (\$370,971.87) but are followed closely by Electronics, Clothing, Home Goods, and Sports, who are within \$24,000 of each other in terms of total revenue.

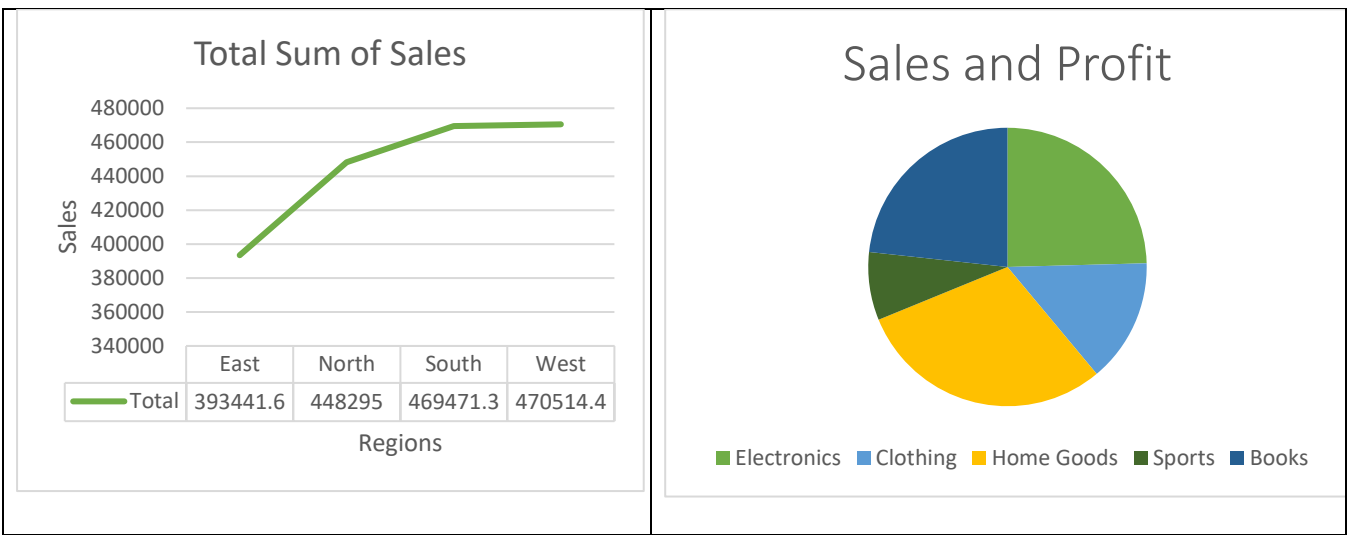
This is healthy as this implies that the risk from sales is distributed well and the company does not rely on any single product type.

The average profit per sale transaction (\$248.70) is relatively consistent across categories indicating solid and consistent margin control. The most crucial management item that comes out of the data quality issues with many zero-dollar sales and entries in wrong categories which needs to be addressed to improve the reporting and analysis reliability.

4. Sales by Category — Chart

The chart below visualizes sales and profit revenue generated by each product category during the 2023 financial year, the quantity category, Total sum of sales and finally pie chart of Sales and profit.





5. Management Recommendations

Based on the findings above, the following three recommendations are proposed to management:

1. Maintain balanced investment across all five product categories.

As the sales performance across Books, Electronics, Clothing, Home Goods and Sports is highly balanced (within a 7% variation), the marketing spends and stock budget in any one product line should not be concentrated, and the brand should continue supporting all five product lines as part of a blended portfolio.

2. Prioritize a data quality audit before the next reporting cycle.

45 transaction entries (.5% of the total records) have a sales amount of \$0.00, yet appear to have valid quantity and profit information, while an additional 7 transactions contain erroneous category information. A person on the management team should be designated to investigate and correct this information at the time of data entry / Point of Sale; this currently depresses true sales results and may interfere with the trending of results in the future.

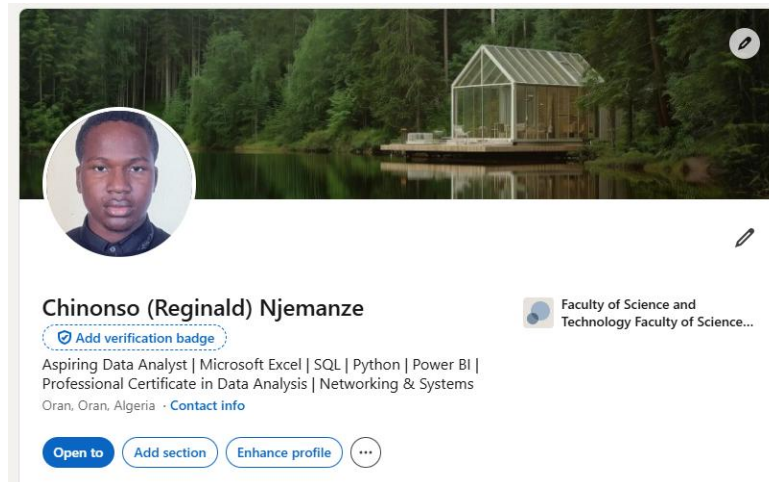
3. Promote Books and Electronics through targeted marketing.

Top two category-totaled sales amount was Books (\$370,971.87), followed closely by Electronics (\$364,090.41). For the 2 biggest revenue driver category total sales, a slightly small, focused campaign effort in either direction should allow lead of both to start translate to a more definite lead, and yet the growth remains, but not at the detriment of 3rd, 4th and 5th largest sales category totals.

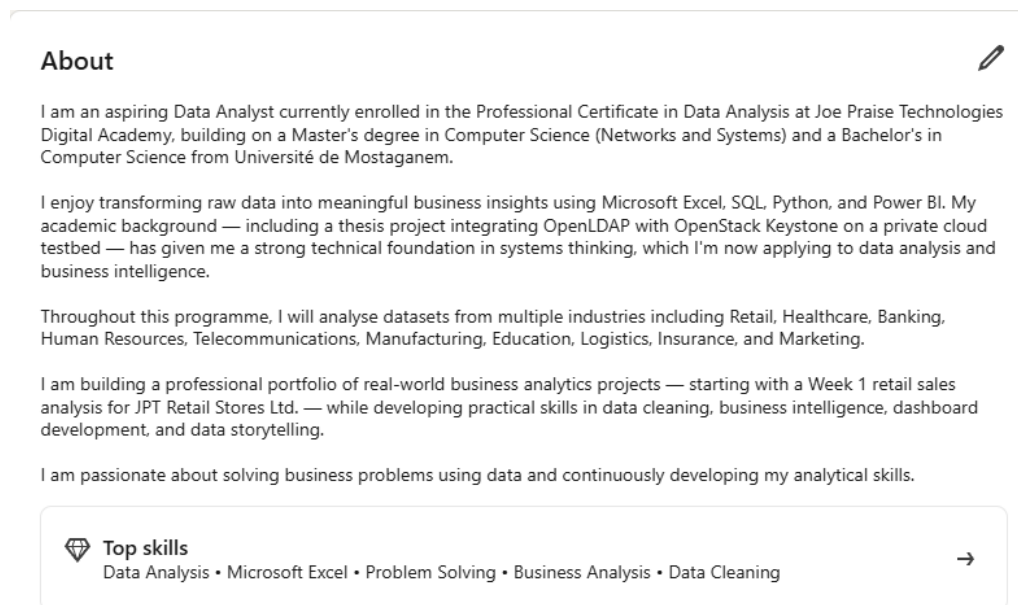
6. Screenshots

The screenshots of the week 1 project.

1. LinkedIn Profile.



2. About Section.



3. Week 1 Post Screenshot.



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Just finished Week 1 of my Professional Certificate in Data Analysis with Joe Praise Technologies Limited! 🚀

This week, I had the opportunity to act as Junior Data Analyst for a fictional retail company, analyse a year's worth of their sales data with Microsoft Excel and the key takeaways included:

- ✅ Data cleaning & preparation for analysis
- ✅ Calculation and interpretation of business metrics (sums, averages, min/max)
- ✅ Identification of data quality issues (\$0.00 "sales" and mislabeled categories, wow!)
- ✅ Data visualization and reporting: turning raw numbers into actionable insights.

I leveraged functions like SUM, AVERAGE, MAX, MIN, IF, and pivot tables to break down sales by category, region, and month.

Key business insights derived:

- 🔥 Total sales: \$1.79M from 1,825 transactions
- 🔥 Average profit margin: Consistent at around 25%
- 🔥 Product sales are well-distributed, but one region is underperforming by roughly 16%

It's reinforced that excellent data analysis is as much about questioning the 'why' as it is about the 'what'.

Ready to tackle Week 2 and continue growing my Data Analyst portfolio over the next 11 weeks!

#DataAnalysis #MicrosoftExcel #LearningJourney #JoePraiseTechnologies #CareerDevelopment



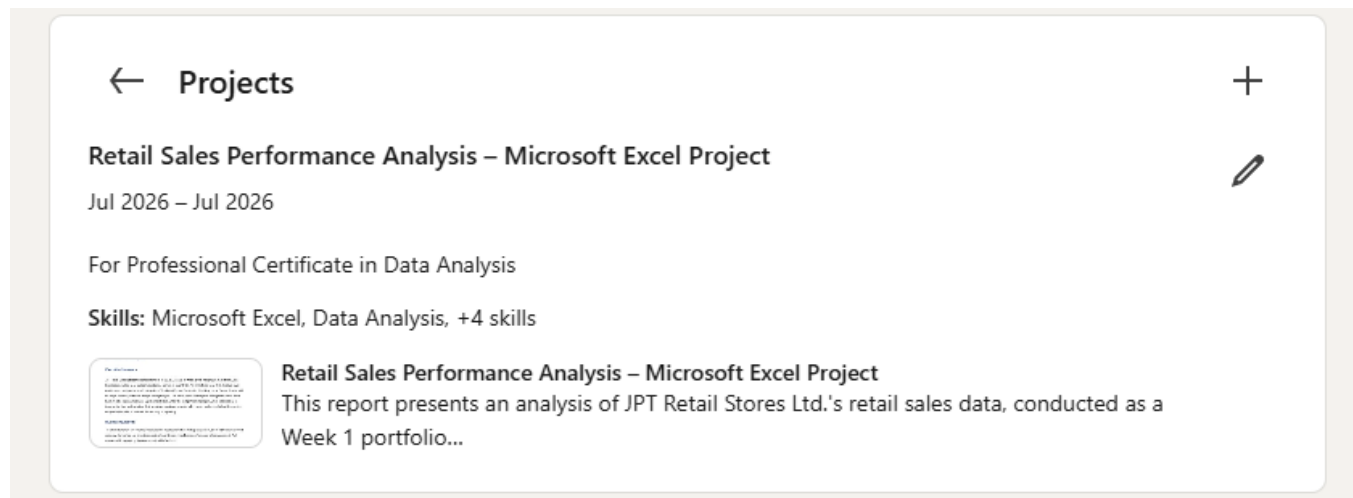





26 impressions

View analytics

4 Headline Screenshot.



5 LinkedIn Url

<https://www.linkedin.com/in/chinonso-njemanze-18a3a640b>

7. Conclusion

JPT Retail Stores Ltd. demonstrates a financially healthy and well-diversified sales base, with total 2023 sales of \$1,787,419.92 and a consistent ~25% profit margin across all product categories. The principal opportunity for improvement lies not in sales strategy but in data governance: resolving the identified zero-value and mislabeled entries will strengthen the reliability of all future business reporting. JPT Retail Stores Ltd. Shows financially sound with well-distributed sales, with the total sales of \$1,787,419.92 during the 2023 period and maintained profitability on each product category at ~25% each year. Sales approach cannot improve but an investment into a clean dataset the only chance that business reporting can stand on it, and fixing zero value and misleading sales report.